

- 7 (a) each line corresponds to a (specific) photon energy ..... B1  
 photon emitted when electron changes its energy level ..... B1  
 discrete energy changes so discrete levels ..... B1 [3]
- (b) (i)  $E = hc / \lambda$  ...(*allow ratio ideas*) ..... C1  
 $= (6.63 \times 10^{-34} \times 3.0 \times 10^8) / (486 \times 10^{-9})$   
 $= 4.09 \times 10^{-19} \text{ J}$  ..... A1 [2]
- (ii) four transitions to/from  $-5.45 \times 10^{-19} \text{ J}$  level ..... B1  
 all transitions shown from higher to lower energy (level) ..... B1 [2]

[Total: 7]

- 8 (a) (constant) probability of decay ..... M1